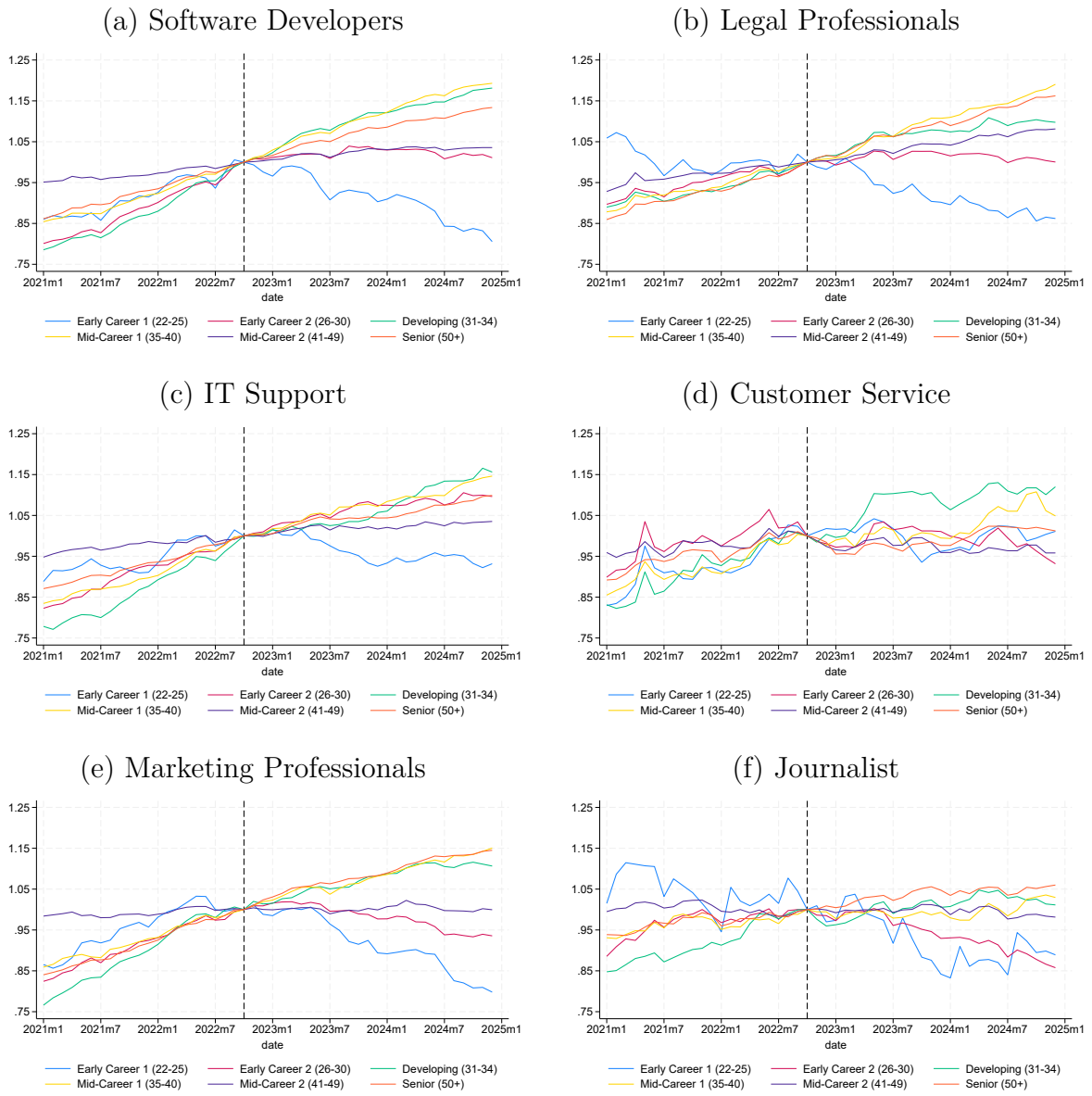
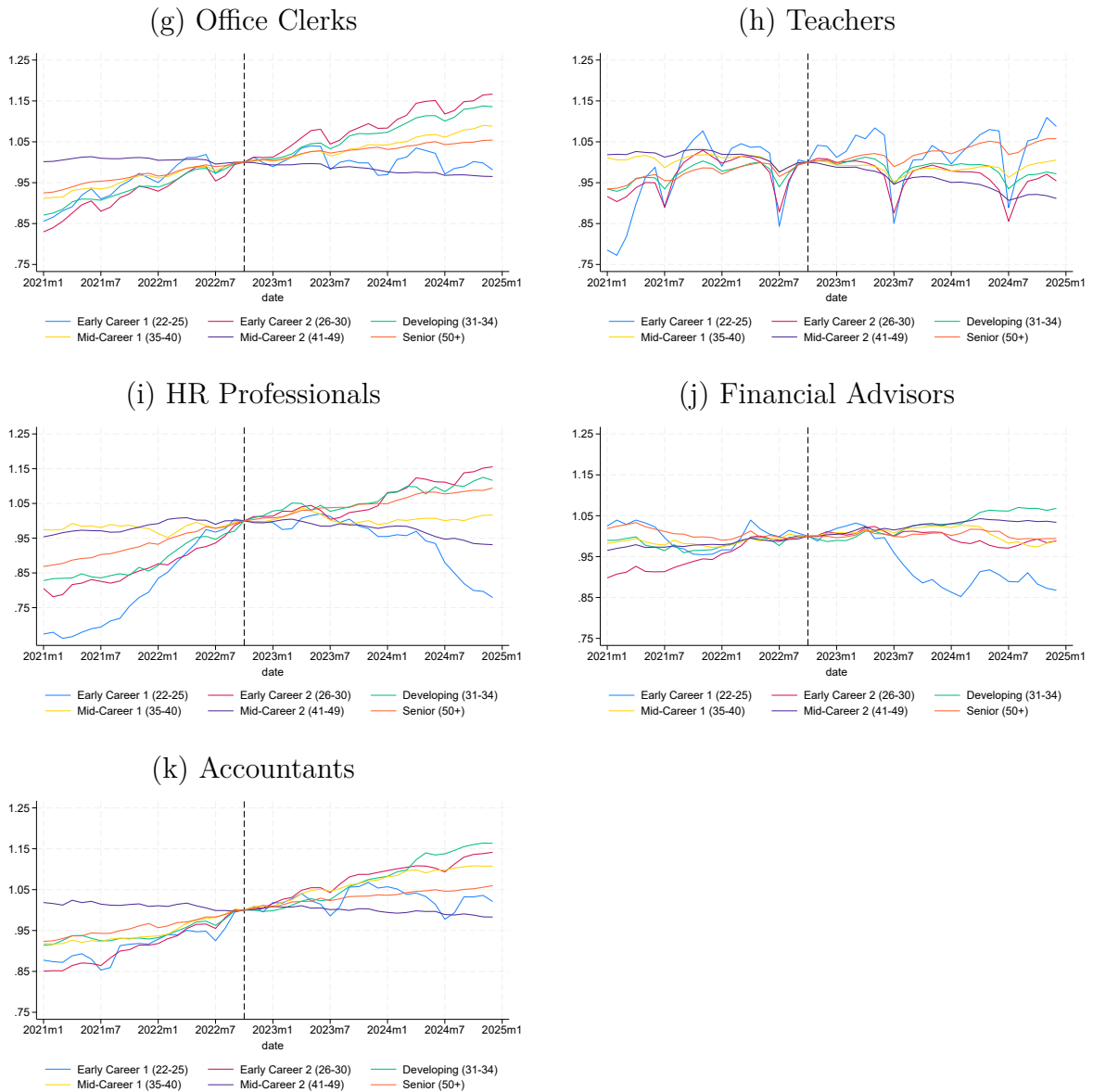


Figure G.2: Employment Trends by Occupation and Age Groups



Notes: This figure is continued on the next page.

Figure G.2: Employment Trends by Occupation and Age Groups (Continued)



Notes: Figure G.2 shows the employment trends for each of the occupations and age groups, indexing these to October 2022. We follow Brynjolfsson, Chandar and Chen (2025) in defining the age groups: early career 1 (ages 22–25), early career 2 (26–30), developing (31–34), mid-career 1 (35–40), mid-career 2 (41–49), and senior (50+). We interpolate occupational codes within job-spells (worker-workplace pairs) to minimize measurement error resulting from the reclassification of occupational codes (Groes, Kircher and Manovskii, 2015; Humlum, 2021). *Sample:* All registry data.

H Estimation Procedures

H.1 Adopter Propensity

- Appendix H.1 details the estimation of these adopter propensities, which our economic framework in Section A.2 microfound as workers' latent willingness to use AI chatbots.

We measure adopter propensity of job creation/destruction as the predicted adoption rate among hires/separations at workplace j in month t based solely on workers' predetermined individual characteristics:

1. Estimate adoption propensity:

$$\text{Adopt}_i = \gamma' X_i + \beta' \text{EmployerInitiatives}_i + \varepsilon_i, \quad (73)$$

where X_i includes predetermined worker characteristics (occupation, gender, age, experience, and earnings)

2. Among workplaces j with positive hires/separations in month t , measure the adopter propensity of their job creation/destruction as:

$$\text{JobCreationAdopterPropensity}_{jt} = \hat{\gamma}' \mathbb{E}[X_i \mid j_t(i) = j, j_{t-1}(i) \neq j] \quad (74)$$

$$\text{JobDestructionAdopterPropensity}_{jt} = \hat{\gamma}' \mathbb{E}[X_i \mid j_t(i) \neq j, j_{t-1}(i) = j], \quad (75)$$

where $j_t(i)$ is the workplace of worker i in month t .

Workers' adopter propensities $\hat{\gamma}' X_i$ correspond to their willingness to adopt AI chatbots in our theoretical framework (Section A).

H.2 Empirical Bayes Shrinkage

As a robustness check, we estimate workplace rates of encouragement using Empirical Bayes shrinkage with a Beta-Binomial model; see Walters (2024) for a detailed introduction

to Empirical Bayes methods. The shrinkage is performed separately for each occupation, allowing underlying encouragement rates to vary systematically across occupations.

We assume the adoption rate at each workplace, p_i , follows a Beta prior:

$$x_i \mid p_i \sim \text{Binomial}(n_i, p_i), \quad p_i \sim \text{Beta}(\alpha_0, \beta_0). \quad (76)$$

The Beta prior captures workplace-level variation. We estimate α_0, β_0 via Method of Moments, matching the Beta distribution's first two moments to observed data:

$$\bar{p} = \frac{1}{m} \sum_{i=1}^m \frac{x_i}{n_i}, \quad s^2 = \frac{1}{m} \sum_{i=1}^m \left(\frac{x_i}{n_i} - \bar{p} \right)^2. \quad (77)$$

From the Beta mean and variance formulas:

$$\alpha_0 = \frac{\bar{p}(1 - \bar{p})}{s^2} - 1, \quad \beta_0 = \alpha_0 \frac{1 - \bar{p}}{\bar{p}}.$$

With these, we compute the posterior mean:

$$\mathbb{E}[p_i \mid x_i] = \frac{\alpha_0 + x_i}{\alpha_0 + \beta_0 + n_i}.$$

This shrinks estimates toward the overall mean, especially for small n_i .

H.2.1 Coworker Encouragement Rates

Figure H.1 compares the raw and adjusted distributions of coworker encouragement rates, while Table H.1 presents summary statistics for the adjusted rates of workplaces. The typical standard deviation within occupations is 17 percentage points. Importantly, our results in Section E.5.2 remain robust when using the raw coworker encouragement rates instead.